

Innovation

A Peek Into Some

OPEN SOURCE-PACKED INNOVATIONS

For some, innovation is fun. They like to create new products or do things in new ways, just for kicks. For others, innovation is a cause, often with a social twist. They look at it as a means to provide for the poor, or to improve the lives of specially-challenged people. This month's Innovation page features both kinds—the common thread that binds them is open source.

A robot-like light that searches and illuminates dark spots

<http://www.livingthings.cz/tvor/>



Designed by Jindrich Vodicka, Tvor is an Arduino-based lamp that keeps watching a room to find the darkest spot, and rushes to illuminate it. It first starts by turning on the spot, monitoring the light in all directions. Then, once it identifies the darker area, it dims its own light and moves in that direction to illuminate it, and then starts the search all over again. It is a non-stop process, as lighting up one area will darken another. Incidentally, it is not of much practical use, because you might not really need light at the darkest spot, if you are not anywhere close to it. Yet, the Tvor lamp has become quite well-known globally, as it is interesting to watch it at work, and it would make good room or restaurant décor. Tvor has dials that control speed, etc, and it is also capable of avoiding obstructions.

The open twist: Tvor uses three key sensors: a photo resistor to detect the intensity of light from all directions, an ultrasound distance sensor to prevent collisions with

obstructions, and a tilt sensor that stops and rotates Tvor when something prevents the movements of the wheels. The main light is positioned on top like a head, and another light-source is positioned near the neck of the object, which pulses subtly, creating a feeling of heartbeats and breathing. An Arduino Mega circuit board is used to process the data from the sensors, run the motors and actuators, and control the intensity of the light and 'heartbeats'. Arduino is an open source electronics prototyping platform—its hardware designs are open licensed and available online, and so is its software. More information on the materials, parts and designs used to build Tvor is available on the website.

A wrist-worn alternative to cloud computing

<http://stormfly.nowcomputing.com/>

A new device called StormFly can store an entire operating system and data on a small, wrist-worn device. The 16 GB device is USB 3.0 powered, and can be used with most modern computers including Macs. As soon as you plug a StormFly into the USB drive, the computer's

